

Students' Name: _____ Teacher: _____



Saint Ignatius' College, Riverview Mathematics Assessment Task 2018

Year 11
Mathematics
Task 3
PRELIMINARY COURSE EXAMINATION
Date: MONDAY 24 th SEPTEMBER 2018

General Instructions:

- **Reading time: 5 Minutes**
- **Time Allowed: 120 Minutes**
- Write using blue or black pen only
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Teachers:

Mr R Maxwell	REM
Mr P Collins	PPC
Dr M Furtado	MXF
Ms F Yates	FEY
Ms H Haxby	HBH
Ms K Mullan	KXM

Topics Examined:

- Question 1 - Basic Arithmetic, Algebra and Surds
- Question 2 - Equations
- Question 3 - Geometry
- Question 4 - Functions and Graphs
- Question 5 - Trigonometry
- Question 6 - Linear Functions
- Question 7 - Introduction to Calculus

Total

84 Marks

Question 1 Basic Arithmetic, Algebra and Surds**Total 12 Marks*****Please use a separate writing booklet.***

- a. Find the value of $\frac{2\pi^2 + 3.14}{\sqrt{\pi}}$ correct to 3 significant figures. **1**
- b. Simplify $\sqrt{18} - \sqrt{8}$, in simplest surd form. **1**
- c. If $V = \frac{4}{3} \pi r^3$ find r , when $V=20.6$, to 2 significant figures. **2**
- d. Simplify $\frac{a-4}{a^3-64}$. **2**
- e. Solve $2 - 3x \leq 8$, and graph the solution on a number line. **2**
- f. Factorise $25 - (x-1)^2$. **2**
- g. If $x = \sqrt{5} + 2$, find b when $x + \frac{1}{x} = b\sqrt{5}$. **2**

Question 2**Equations****Total 12 Marks*****Please use a separate writing booklet.***

a. Solve simultaneously $3x + 2y = 7$ and $3x - 2y = 11$. 2

b. Solve the equation $\frac{x+3}{2} - \frac{2x-1}{5} = 1$. 2

c. Solve $2x^2 + 7x + 3 = 0$. 2

d. Solve $2^{3x-1} = \frac{1}{4}$. 2

e. Solve $(x-3)(x-4) \geq 20$. 2

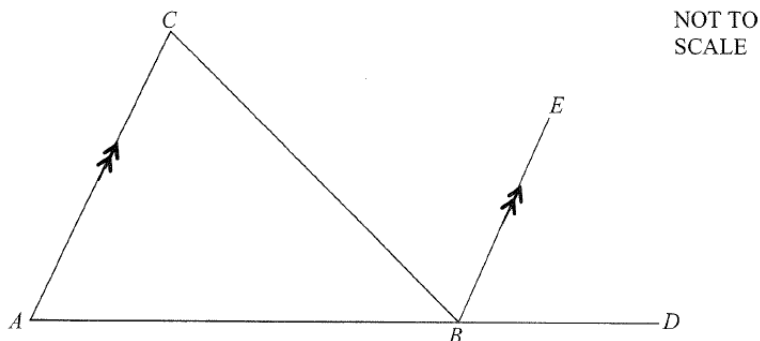
f. Solve $|2x-3| = 3x-1$. 2

Question 3**Geometry****Total 12 Marks**

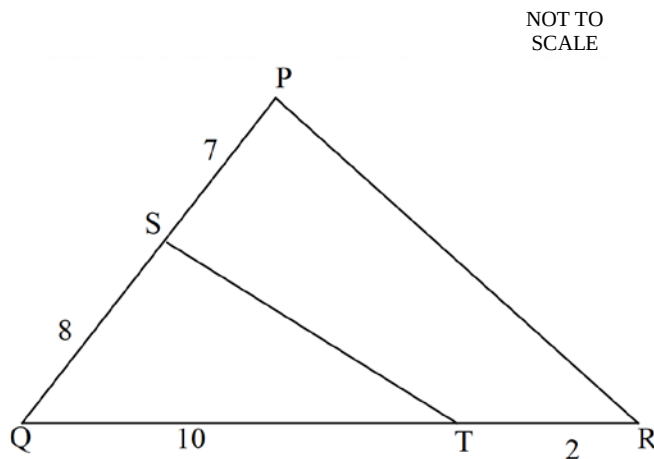
Please use a separate writing booklet.

- a. A regular polygon is such that the size of each interior angle is 9 times the size of each exterior angle. How many sides does the polygon have? 2

- b. In the diagram, the side AB of $\triangle ABC$ is produced to D . BE is parallel to AC and BE bisects $\angle CBD$. Copy the diagram and show that $AB = BC$.



- (i) Copy the diagram on your answer booklet 1
- (ii) Show that $AB = BC$. (Giving Reasons) 3
- (iii) If $\angle EBD = 50^\circ$. Find the measure of $\angle CBA$. 1
- c. Given the diagram below:



- (i) Prove that triangle QST is similar to triangle QPR. 3
- (ii) Hence find the length of PR, given that $ST = 6$. 2

Question 4**Functions and Graphs****Total 12 Marks*****Please use a separate writing booklet.***

- a. Find the $\lim_{x \rightarrow 2} \frac{x^4 - 16}{4x - 8}$. 2
- b. Consider the function $f(x) = 2x^4 + 3x^2 + 4$. Show algebraically that the function $f(x)$ is either odd, even or neither. 2
- c. Sketch the graph $y = |x + 4|$.
Showing all intercepts on the x and y axis. 2
- d. (i) Sketch the function $y = -\sqrt{9 - x^2}$.
Showing all intercepts on the x and y axis. 2
- (ii) State the range of this function. 1
- e. (i) Sketch the graph of the function $f(x) = \sqrt{x} - 2$.
Showing all intercepts on the x and y axis. 2
- (ii) State the domain of this function. 1

Question 5**Trigonometry****Total 12 Marks***Please use a separate writing booklet.*

a. Show that $\frac{\cot\theta}{\operatorname{cosec}\theta} = \cos\theta$. 2

b. Show that $\cos\theta + \tan\theta \sin\theta = \sec\theta$. 2

c. Solve $\sin^2\theta - 1 = 0$. $0^\circ \leq \theta \leq 360^\circ$. 2

d. Given $\sin\theta = -\sin 50^\circ$ for $0^\circ \leq \theta \leq 360^\circ$. Find the values of θ . 2

e. In $\triangle ABC$, $BC = 19\text{cm}$, $AC = 22\text{cm}$ and $\angle ABC = 55^\circ$.

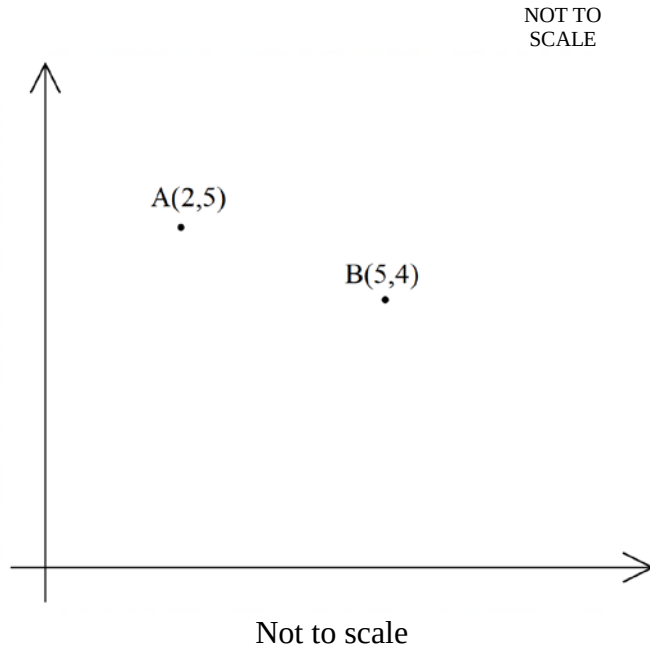
(i) Find the size of $\angle CAB$ to the nearest degree. 2

(ii) Hence find the area of $\triangle ABC$, to the nearest square centimetre. 2

Question 6**Linear Functions****Total 12 Marks**

Please use a separate writing booklet.

- a. The diagram below shows the points $A(2,5)$ and $B(5,4)$.



- | | |
|--|---|
| (i) Find the coordinates of M the mid-point of AB . | 1 |
| (ii) Show that the equation of the line AB is $x + 3y = 17$. | 2 |
| (iii) Show that the equation of the perpendicular bisector of AB is $3x - y = 6$. | 2 |
| (iv) The perpendicular bisector of AB cuts the x -axis at C .
Find the coordinates of C . | 1 |
| (v) Find the area of $\triangle ABC$. | 2 |
- b. What is the value of k such that the lines $k^2x - 4y = 0$ and $y = \frac{2}{k}x$,
are perpendicular to one another? 2
- c. Find the exact value of y , given that the points $(2, 3)$ and $(1, y)$
make an angle of 60° with the x axis. 2

Question 7**Introduction to Calculus****Total 12 Marks*****Please use a separate writing booklet.***

a. Find the derivatives of the following functions:

(i) $y = \frac{1}{3}x^{\frac{3}{2}}$ **1**

(ii) $y = (3x + 1)^4$ **1**

(iii) $y = x^2(4x + 1)^3$. (Answer to be fully factorised and simplified) **2**

b. Find the coordinates of the points at which the curve $y = x^4 + 1$,
has a tangent with gradient 4? **2**

c. Using first principles **ONLY** differentiate the function $f(x) = 2x^2 - x$. **3**

d. (i) Find the gradient of the tangent to the curve $y = \frac{3x}{x-2}$ at the point on the curve
where $x = 3$. **1**

(ii) Hence find the equation of the normal to the given curve. **2**

SOLUTIONS



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Total

84 Marks

Solutions

Comments

Question 1

(a) $12 \cdot 9$

(b) $3\sqrt{2} - 2\sqrt{2} = \sqrt{2}$

(c) $\pi r^3 = 20 \cdot 6$

$$r^3 = \frac{20 \cdot 6}{\pi}$$

$$r = 1.9 \text{ units.}$$

(d) $\frac{(a-4)}{(a-4)(a^2+4a+16)}$

$$= \frac{1}{(a^2+4a+16)}$$

(e) $2x^2+7x+3=0$

$$(2x+1)(x+3)=0 \quad \therefore x = -\frac{1}{2}, x = -3$$

(f) $5^2 - (x-1)^2 = (5-x+1)(5+x-1)$

$$= (6-x)(4+x)$$

$$\text{or } = -(x+4)(x-6)$$

(g) $\sqrt{5}+2 + \frac{1}{\sqrt{5}+2} = \sqrt{5}+2 + \frac{(\sqrt{5}-2)}{(\sqrt{5}+2)(\sqrt{5}-2)}$

$$= \frac{\sqrt{5}+2+\sqrt{5}-2}{5-4} = 2\sqrt{5} = b\sqrt{5}$$

$$\therefore b = 2$$

Solutions

Comments

Question 2

$$(a) \quad 3x + 2y = 7 \quad -1$$

$$+ \quad 3x - 2y = 11 \quad -2$$

$$\therefore 6x = 18$$

$$x = 3$$

$$\text{Sub } x=3 \text{ in 1: } 3(3) + 2y = 7$$

$$2y = -2 \quad y = -1$$

$$\therefore x = 3, y = -1$$

$$(b) \quad 5(x+3) - 2(2x-1) = 10$$

$$5x + 15 - 4x + 2 = 10$$

$$x + 17 = 10$$

$$x = -7$$

$$(c) \quad \frac{3x-1}{2} = 2 \quad -2$$

$$\therefore 3x-1 = -2$$

$$3x = -1$$

$$x = -\frac{1}{3}$$

$$(d) \quad 2x-3 = 3x-1 \text{ and } 2x-3 = -3x+1$$

$$-x = 2$$

$$5x = 4$$

$$x = -2$$

$$x = \frac{4}{5}$$

$$\text{Check: } 3(-2)-1 = -7 < 0 \quad 3\left(\frac{4}{5}\right)-1 = \frac{7}{5} > 0$$

$\therefore$ Not a Soln.

$$\therefore x = \frac{4}{5}$$

Solutions

Comments

Question 2 (cont)

(e) $2 - 3x \leq 8$

$$-3x \leq 6$$

$$3x \geq -6$$

$$x \geq -2$$

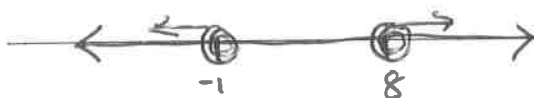


(f) $(x-3)(x-4) \geq 20$

$$x^2 - 7x + 12 - 20 \geq 0$$

$$x^2 - 7x - 8 \geq 0$$

$$(x-8)(x+1) \geq 0$$



$$\therefore x \leq -1$$

$$x \geq 8$$

Solutions

Comments

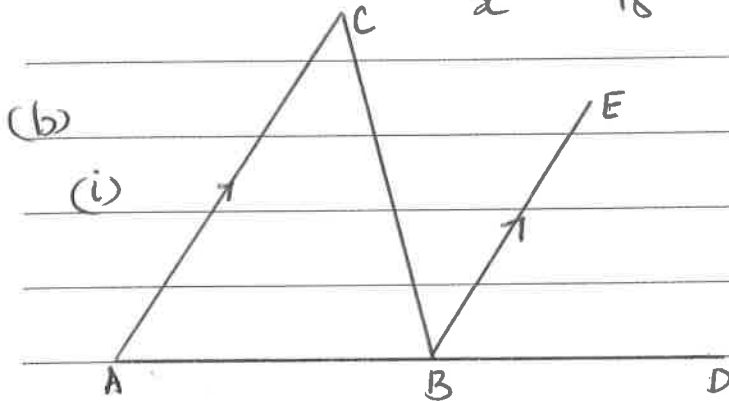
Question 3

(a) $\frac{9x}{x}$

$$\therefore 10x = 180^\circ$$

$$x = 18^\circ = \text{Ext. } \angle A$$

$$\therefore \text{No. of sides} = \frac{360}{x} = \frac{360}{18} = 20$$



(ii)

Reason

1. $\angle DBE = \angle EBC$ (Given) BE bisects $\angle CBD$

2. $\angle BCA = \angle EBC$ Alternate $\angle$'s on parallel lines

AC and BE are equal.

3. $\angle BAC = \angle DBE$ Corresponding $\angle$'s on parallel lines

AC and BE are equal.

$$\therefore \angle BCA = \angle BAC$$

$$\therefore AB = BC \text{ (Sides opposite equal } \angle \text{'s are equal)} \\ \text{in } \triangle ABC$$

(iii) $\angle EBD = 50^\circ$

$$\therefore \angle CBA = 80 \text{ (No Reason Reqd.)}$$

Solutions

Comments

Question 3 (cont)

(c) (i) Prove $\triangle QST \parallel \triangle QPR$.

1. $\angle SQT = \angle PQR$ (A) Common Angle

2. $\frac{QT}{QP} = \frac{QS}{QR}$ as

$$\frac{10}{15} = \frac{8}{12} = \frac{2}{3} \therefore$$

$\therefore$ Sides Corresponding to the
2 $\triangle$'s are proportional.

3. $\therefore \triangle QST \parallel \triangle QRP$

(Two triangles are similar if the
ratios of two pairs of corresponding
sides are equal and the angles
included by these sides are equal)

(ii) $\therefore$ The ~~the~~ third sides are
also proportional. i.e.

$$\frac{QT}{QP} = \frac{QS}{QR} = \frac{ST}{PR} = \frac{2}{3}$$

$\therefore$ Given $ST = 6$

$$\frac{ST}{PR} = \frac{2}{3}$$

$$PR = \frac{6 \times 3}{2} = 9$$

Solutions

Comments

Question 4

$$(a) \lim_{x \rightarrow 2} \frac{x^4 - 16}{4x - 8}$$

$$= \lim_{x \rightarrow 2} \frac{(x^2 - 4)(x^2 + 4)}{4(x - 2)}$$

$$= \lim_{x \rightarrow 2} \frac{\cancel{(x-2)}(x+2)(x^2+4)}{4\cancel{(x-2)}}$$

$$= \frac{(2+2)(4+4)}{4}$$

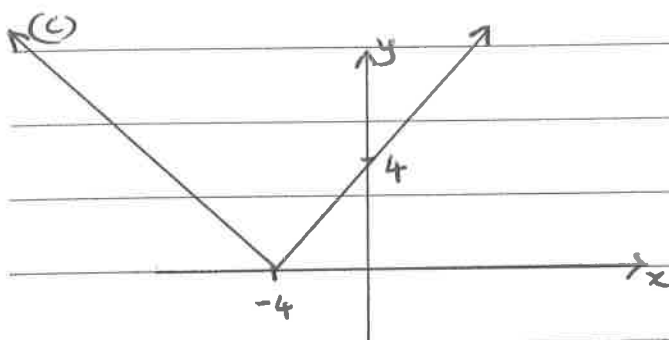
$$= \frac{4 \times 8}{4} = 8$$

$$(b) f(x) = 2x^4 + 3x^2 + 4$$

$$f(-x) = 2(-x)^4 + 3(-x)^2 + 4$$

$$= 2x^4 + 3x^2 + 4$$

$$= f(x) \quad \therefore \text{EVEN fn.}$$

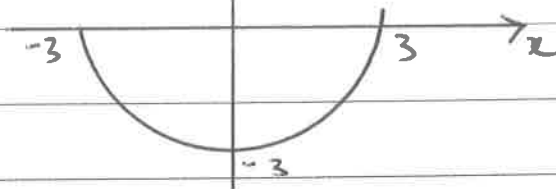


Solutions

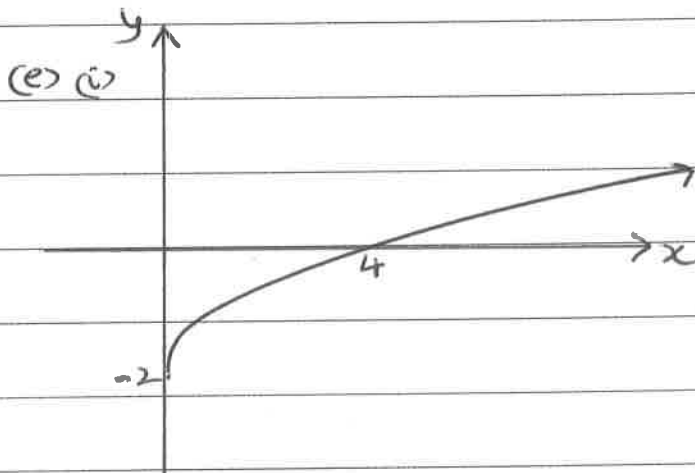
Comments

Question 4 (cont)

(d) (i)



(ii) Range: $-3 \leq y \leq 0$



(iv) Domain: $x \geq 0$

Solutions

Comments

Question 5:

$$(a) \frac{\cos \theta}{\frac{\sin \theta}{\frac{1}{\sin \theta}}} = \frac{\cos \theta}{\sin \theta} \times \frac{\sin \theta}{1} = \cos \theta$$

$$(b) \cos \theta + \frac{\sin \theta}{\cos \theta} \times \sin \theta$$

$$\frac{\cos^2 \theta + \sin^2 \theta}{\cos \theta} = \frac{1}{\cos \theta} = \sec \theta$$

$$(c) \sin^2 x - 1 = 0$$

$$(\sin x - 1)(\sin x + 1) = 0$$

$$\therefore \sin x = 1 \quad \sin x = -1$$

$$x = 90^\circ \quad x = 270^\circ$$

$$(d) \sin \theta = -\sin 50$$

		$\theta = 50^\circ$
S	A	
T	C	
$180 + \theta$	$360 - \theta$	
✓	✓	

$$\therefore \theta = 180 + 50 = 230^\circ$$

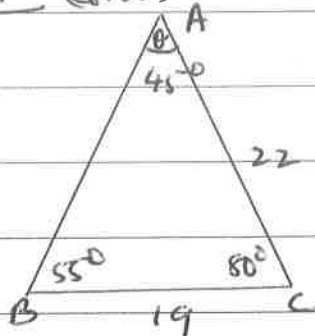
$$\text{or } \theta = 360 - 50 = 310^\circ$$

Solutions

Comments

Question 5 (cont)

(c)



$$(i) \therefore \frac{\sin \theta}{19} = \frac{\sin 55^\circ}{22}$$

$$\sin \theta = \frac{19 \times \sin 55^\circ}{22} = 0.707 \dots$$

$$\theta = 45^\circ \text{ (nearest degree)}$$

(ii) Area of $\triangle ABC$

$$= \frac{1}{2} ab \sin \theta$$

$$= \frac{1}{2} \times 19 \times 22 \times \sin 80^\circ$$

$$= 205.82 \dots$$

$$= 206 \text{ sq. cms.}$$

Solutions

Comments

Question 6

$$(i) M = \left[\frac{2+5}{2}, \frac{5+4}{2} \right] = \left(\frac{7}{2}, \frac{9}{2} \right)$$

$$(ii) m_{AB} = \frac{4-5}{5-2} = -\frac{1}{3}$$

$\therefore$ Equation of AB

$$y-5 = -\frac{1}{3}(x-2)$$

$$3y-15 = -x+2$$

$$x+3y = 17 \quad \text{or} \quad x+3y-17 = 0$$

(iii) Equation of $\perp$ ar bisector of AB

$$m_{AB} = -\frac{1}{3} \quad \therefore m \text{ of } \perp \text{ is } 3$$

$\therefore$ Equation

$$y - \frac{9}{2} = 3\left(x - \frac{7}{2}\right)$$

$$2y-9 = 3(2x-7)$$

$$2y-9 = 6x-21$$

$$6x-2y = 12$$

$$3x-y = 6 \quad \text{or} \quad 3x-y-6 = 0$$

$$(iv) y=0 \text{ in } 3x-y-6=0$$

$$\therefore 3x = 6 \quad x = 2 \quad C(2,0)$$

$$(v) \text{ Area of } \triangle ABC = \frac{1}{2} \times b \times h$$

$$d_{MC} = \frac{|x+3y-17|}{\sqrt{1+9}} = \frac{|2-17|}{\sqrt{10}} = \frac{15}{\sqrt{10}}$$

$$AB = \sqrt{(5-2)^2 + (4-5)^2} = \sqrt{10}$$

$$A = \frac{1}{2} \times \frac{15}{\sqrt{10}} \times \sqrt{10} = 7.5 \text{ u}^2$$

Solutions

Comments

Question 6 (cont)

$$(b) \quad m_1 \times m_2 = -1$$

$$m_1 = \frac{k^2}{4} \quad m_2 = \frac{2}{k}$$

$$\therefore \frac{k^2}{4} \times \frac{2}{k} = -1$$

$$\frac{k}{2} = -1$$

$$k = -2$$

$$(c) \quad \tan \theta = \frac{y-3}{1-2}$$

$$\sqrt{3} = \frac{y-3}{-1}$$

$$-\sqrt{3} = y-3$$

$$y = 3 - \sqrt{3}$$

Solutions

Question 7

Comments

$$a) i) y = \frac{1}{3} x^{\frac{3}{2}}$$

$$y' = \frac{1}{3} \times \frac{3}{2} \times x^{\frac{1}{2}}$$

$$= \frac{1}{2} x^{\frac{1}{2}}$$

$$(ii) y = (3x+1)^4$$

$$y' = 4(3x+1)^3 \times 3$$

$$= 12(3x+1)^3$$

$$(iii) y = x^2(4x+1)^3 \quad u = x^2 \quad v = (4x+1)^3$$

$$u' = 2x$$

$$v' = 3 \times 4(4x+1)^2$$

$$u' = 2x + 12(4x+1)^2$$

$$y' = 2x(4x+1)^3 + 12x^2(4x+1)^2$$

$$= 2x(4x+1)^2 [4x+1 + 6x]$$

$$= 2x(4x+1)^2 (10x+1)$$

$$(b) y = x^4 + 1$$

$$y' = 4x^3 = 4$$

$$x^3 = 1$$

$$x = 1, \quad y = 1 + 1 = 2$$

$\therefore$ Coord: (1, 2)

Solutions

Comments

Question 7: Cont.

$$(c) f(x) = 2x^2 - x$$

$$f(x+h) = 2(x+h)^2 - (x+h)$$

$$= 2x^2 + 4xh + 2h^2 - x - h$$

$$f(x+h) - f(x) = 4xh + 2h^2 - h$$

$$\therefore \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x)$$

$$= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 - h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(4x + 2h - 1)}{h}$$

$$\text{Let } h = 0 \quad = 4x + 2h - 1$$

$$= 4x + 0 - 1$$

$$= 4x - 1$$

$$(d)(i) \quad y = \frac{3x}{x-2} \quad u = 3x \quad v = x-2$$

$$u' = 3 \quad v' = 1$$

$$\therefore y' = \frac{3(x-2) - 3x}{(x-2)^2}$$

$$\text{at } x = 3, m_t = \frac{-6}{1} = -6$$

$$m_n = \frac{1}{6}, \quad x = 3, \quad y = 9$$

(ii) $\therefore$ Eqn. of normal:

$$y - 9 = \frac{1}{6}(x - 3)$$

$$6y - 54 = x - 3$$

$$x - 6y + 51 = 0$$

[illegible]